

”Government Spending Multipliers in Good Times and in Bad: Evidence from US Historical Data”

V. Ramey and S. Zubairy (2017)

A replication

Judith Kleman

Advanced Time Series - X/CREST - January 2025

1 Introduction

The literature on fiscal multipliers has undergone numerous developments over time, reflecting the evolution of methodological tools and debates on the effectiveness of fiscal policies. Some of the earliest theoretical work assumed homogeneity of the tax multiplier, whether through a traditional Keynesian approach (suggesting that the multiplier effect of public spending would be constant, depending mainly on the marginal propensity to consume and the effect of savings on the equilibrium), or through a neoclassical approach (in which, under assumptions of rapid adjustment to prevent a lasting under-utilisation of resources, and of monetary policy capable of countering the effects of public spending on aggregate demand, the multipliers are independent of economic conditions). However, this assumption has come up against certain limits. While the conceptualisation of heterogeneity in the impact of fiscal multipliers was already assumed by Keynes (the hypothesis that public spending would have a much greater impact in periods of under-utilisation of resources, such as during a recession or depression, marginal propensity to consume and nominal rigidities increase the effects of fiscal policies when unemployment is high), Samuelson and Tinbergen (works on macroeconomic modelling revealed that public spending during a recession provided a strong economic stimulus due to high unemployment and idle capacity), Hansen (pointing out that the multipliers observed during the Great Depression were higher than in normal times, due to the absence of crowding-out effects), the limits of these traditional models (via the incorporation of nominal rigidities, liquidity constraints leading to hand-to-mouth consumers, or the presence of fiscal-monetary interaction, particularly in periods of zero lower bound (ZLB)) led to an empirical literature developed in the 2010s that tries to measure and identify this hypothetical state-dependence of multipliers (e.g. Fazzari, Morley and Panovska, 2014 with a Bayesian threshold-VAR, using the utilisation capacity rate as the transition variable). Auerbach and Gorodnichenko (2012, 2013), using the Smooth Transition VAR, introduced a method to systematically and robustly differentiate multipliers during recessions and expansions. The multipliers were around twice as high in recessions (1.5 to 2.5) as in expansions (0.5). The article of interest here, by Valerie Ramey and Sarah Zubairy (2017) is part of this literature.

Their approach is based on a narrative method already proposed by Ramey and Shapiro (1998): the authors used specific episodes exogenous to the economic conditions (for example, variations in military spending) to measure economic responses to exogenous shocks, which led them to find that multipliers varied. However, their analysis did not explicitly differentiate between recession and expansion. It also builds on the method developed by Blanchard and Perotti (2012): the authors use a Structural VAR model with a short-run restriction (where government spending is predetermined within the quarter and thus not endogenous to economic growth) to estimate multipliers by, had paved the way for empirical analyses showing that multipliers could be higher in recession, although not differentiating economic states. In addition to studying the case of differentiation according to periods of recession and expansion, they are also based on a literature specifically studying multipliers in periods of zero lower bound (*inter alia* Christiano, Eichenbaum & Rebelo, 2011; Eggertsson, 2011; Crafts & Mills, 2013).

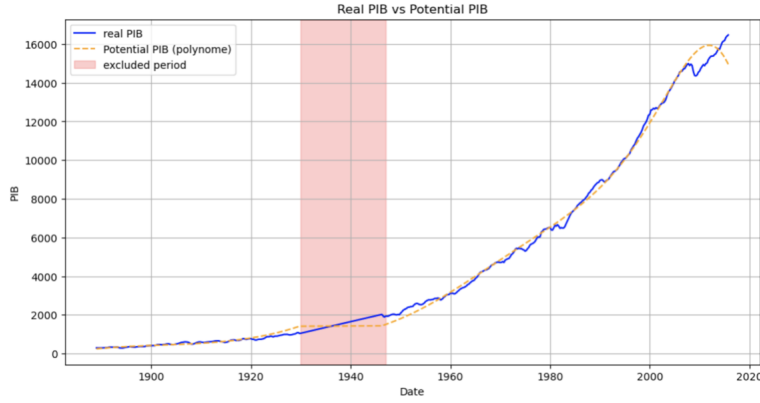


Figure 1: Computation of Potential GDP

Ramey and Zubairy make a number of criticisms of the work of Auerbach and Gorodnichenko concerning the amplitude of multipliers in the event of a recession. In particular, they criticise the simplified assumptions about economic state transitions. Indeed, to construct their impulse responses, the latter assume that the economy remains in its initial state throughout the analysis horizon. According to Ramey and Zubairy, this assumption is unrealistic, because recessions have a limited duration, and the economy often quickly returns to a state of expansion. By artificially maintaining the economy in a constant state, multipliers can be overestimated, particularly during recessions. This criticism leads the authors to use an alternative approach based on (non-linear) Local Projections à la Jordà (2005), which does not require explicit assumptions about the duration of economic states. Ramey and Zubairy also point out that the multipliers calculated by Auerbach and Gorodnichenko can vary significantly depending on the historical period studied. In particular, they propose to use a longer sample that goes back to 1889 (and includes periods such as the Great Depression and the Second World War), leading them to find much more modest multipliers, even in periods of recession.

Ramey and Zubairy also offer a specific criticism of the work on the amplitude of multipliers during the zero lower bound period: faced with the work of Christiano, Eichenbaum and Rebelo (2011) and Eggertsson (2011) predicting multipliers greater than 2 during the ZLB period, they highlight the extremely restrictive assumptions (of no crowding out, perfect anticipation of economic agents concerning the duration of the ZLB period, and perfect coordination between fiscal and monetary policy) used by these DSGE models. They also reformulate the same measure concerning the fact that the overestimated persistence of the ZLBs would artificially amplify the estimators.

The aim of this paper is to replicate the method used by Ramey and Zubairy to estimate tax multipliers, making a specific distinction between the cases of extension and recession, and then zero lower bound. It will also provide a detailed and constructive discussion of their methods and results.

2 Descriptive statistics

The data used in this homework come from the online Appendix to the article. I was unable to find all the data on my own, particularly for certain pre-47 historical data, or for the "news shocks". However, a code has been set up to recalculate the data and variables created by the authors.

2.1 Potential GDP

The authors use the Gordon Krenn method to estimate potential GDP, in order to reconstruct economic series over long historical periods. Potential GDP is estimated by fitting a polynomial of degree 6 to GDP. This calculation of potential GDP is then used to normalise all the data used in the regressions. This is justified by the need to neutralise the effects of long-term economic growth (over long periods,

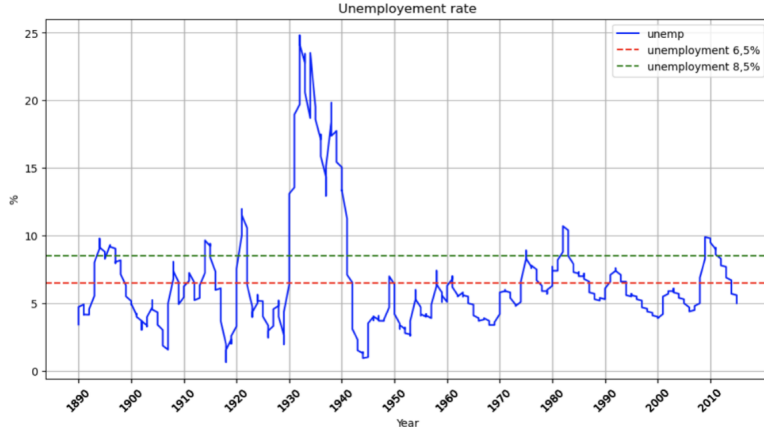


Figure 2: Two unemployment rates as mesure of slack

real GDP tends to grow structurally as a result of technological progress, capital accumulation and demographic growth. Without this division, estimates could be biased by periods when real GDP is exceptionally low or high due to transitory shocks (such as recessions or wars). Normalisation by potential GDP filters out these effects and concen tre the analysis on structural variations. The calculation of potential GDP is based on the period from 1889 to 2015, but excludes the period from 1930 to 1946, which includes the Great Depression and the Second World War. On the one hand, during the Great Depression there was a dramatic fall in output and income, accompanied by high unemployment: real GDP was well below its potential due to the under-utilisation of resources. On the other hand, a period of massive mobilisation of resources, with economic output influenced mainly by the war effort and not by normal market forces, occurred during the Second World War: real GDP then probably exceeded 'natural' economic potential as a result of mobilisation policies and massive public spending. The exclusion of this specific period is particularly justified when calculating potential GDP.

2.2 Slack periods selection

The main recession periods are constructed from a threlshold for a certain unemployment rate. The rate chosen is 6.5%, in line with historical and methodological considerations, as an indicator of under-utilization of resources. Firstly, the choice of this threshold can be explained by the fact that it became a benchmark for estimating slack periods under Ben Bernanke's presidency¹. This was a period of zero lower bound, which is why it is used in this article.

Figure 3 shows that although negative output gaps and unemployment rates below 6.5% are often correlated, this is not always the case. In the 1950s and 1960s, some periods had a moderately negative output gap, without crossing the 6.5% unemployment threshold. Conversely, we can see that in the 1970s, some periods are characterised by a negative output gap, without having an unemployment rate above 6.5%.

Furthermore, it can be considered a robust alternative to identifying slack via the level of natural unemployment. This is because natural unemployment evolves over time as a function of structural changes in the economy (e.g. technological progress, demographics, labour market policies). ls. These variations make it difficult to use a single or constant rate of natural unemployment to identify the slack.

It should also be noted that extreme slack phenomena can be analysed using a threshold of 8%, in the hope of capturing higher budget multipliers. We can see from Figure 4 that this number of periods is naturally smaller.

¹In 2012, against a backdrop of Zero Lower Bound (ZLB), the Fed, under the chairmanship of Ben Bernanke, announced that it would keep interest rates close to zero as long as the unemployment rate remained above 6.5%, except in the event of inflationary pressures.

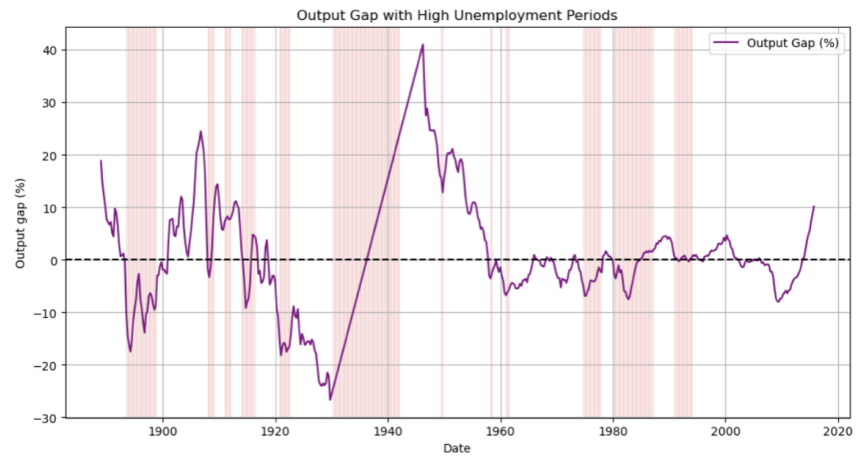


Figure 3: Ouptut gap and slack periods defined by a high unemployment rate

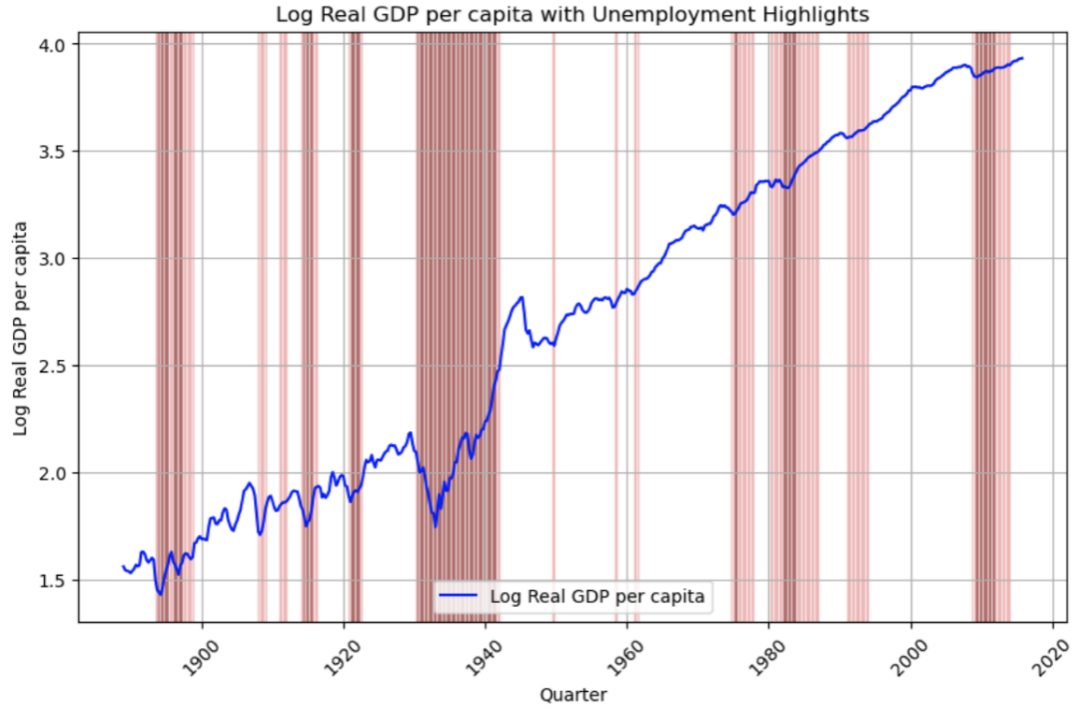


Figure 4: Log real GDP oer capita with unemployment Highlightsn

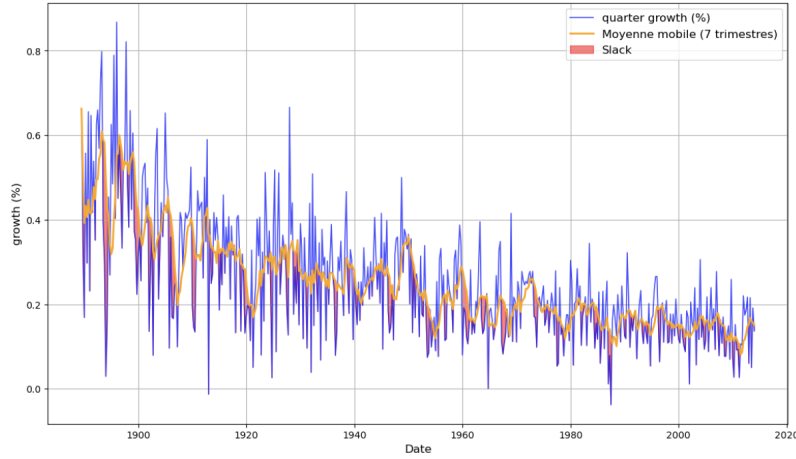


Figure 5: Moving Average of Output Growth Weighting Function

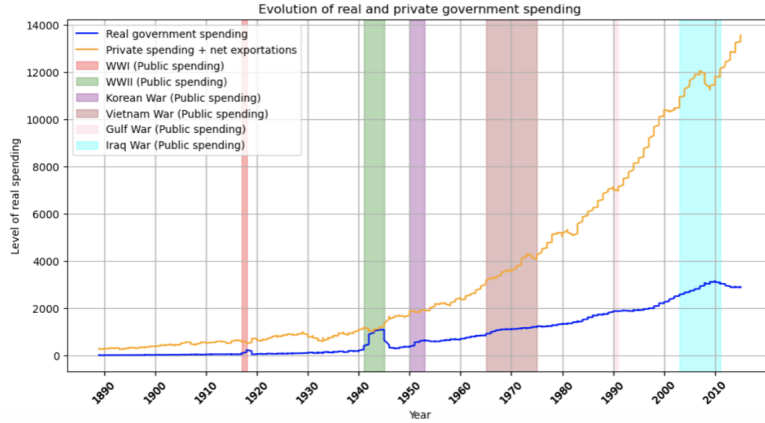


Figure 6: Comparison of real and private spending, accross war periods

Definitions of slack based on a specific unemployment rate, although validated by the NBER, are discrete and do not capture transitions between economic states. One of the solutions used by the authors is to use a weighted function based on the moving average of real GDP growth to define slack periods. This approach captures more gradual variations in economic activity and avoids abrupt breaks in the data. This approach captures more gradual variations in economic activity and avoids abrupt breaks in the data. It also better reflects underlying economic trends over a recent period, eliminating short-term fluctuations or data anomalies. The flip side of this is that moving averages react slowly to abrupt changes, such as sudden recessions or periods of strong growth. Le slack estimé peut sous-estimer ou retarder la détection d'un choc économique important, ce qui rend difficile l'identification des périodes où des instruments comme la politique budgétaire ou monétaire auraient dû réagir. Par ailleurs, la moyenne mobile repose sur une hypothèse implicite que les données sont linéaires ou qu'elles suivent une dynamique stationnaire. Par exemple, si le slack dépend de variables non stationnaires (comme la croissance tendancielle ou des changements structurels dans l'économie), une moyenne mobile risque de capturer des relations incorrectes. Pour toutes ces raisons, le slack de la grande depression pourrait être sous estimé. Elle peut donc manquer des ruptures structurelles, qui peuvent pourtant être motrices dans le déclenchement de récessions. Notons enfin que la fenêtre de lissage, choisie par les autrices (7 quarter) reste arbitraire, et peut induire une perte d'information si elle est trop importante.

Figure 5 shows that there were peaks in public spending during the two world wars (particularly the second) and the Korean War (1950-1953), but also in much smaller amounts during the Vietnam War (1965-1975), the Gulf War (1990-1991) and the Iraq War (2003). The impact on private spending has varied, being moderate for the first world war, much greater for the second, and much lower for

subsequent wars. This phenomenon can be explained, among other things, by the tendency of the American State to mobilise labour and capital to serve the war effort, thereby creating a crowding-out effect. It should be noted that unemployment was artificially reduced by the need for men in the army and in the war industry. Conversely, the two world wars were periods of rationing of consumption, limiting household demand and distorting the relationship between public spending and total output, thus reducing the multiplier, as war spending had a limited effect on future economic productivity. For this reason, the authors suggest excluding at least the Second World War from the multiplier calculations, and potentially including the First World War and the Korean War as well, as the crowding-out effect is less significant but still present. It should be noted that for the Iraq war, the period of rising public spending and falling private spending corresponds mainly to the recession caused by the 2008 crisis.

In addition, it should be noted that a methodological problem is also posed by the period corresponding to the Great Depression: beyond the particular importance of the Slack phenomena compared with other recessions, in the case of the Great Depression it is difficult to identify exogenous fiscal shocks, as the New Deal was a direct response to economic conditions, and increases in public spending were not always clearly announced or anticipated. It should also be noted that the Great Depression was marked by major structural changes, which can distort the dynamics of the budgetary response².

3 Methodology

To address the question of the hypothetical state-dependence of government spending multipliers (e.g. when there is high economic slack or when interest rates are constrained by the zero lower bound (ZLB)), the authors construct a comprehensive US quarterly historical dataset from 1889 to 2015, which includes episodes of large government spending shocks, such as wars and recessions. The methodological approach employed focuses on the identification of exogenous fiscal shocks using narrative methods (à la Romer & Romer 1983), combined with Jordà's (2005) econometric method of Local Projections to generate Impulse Responses (IRFs), using the shocks as exogenous instruments. This allows robust estimation of dynamic impulse response functions and state-dependent multipliers.

3.1 Narrative method and "defense news spending shocks"

The narrative method is based on the identification of public spending shocks from specific historical events, in particular announcements relating to military spending. Unlike traditional econometric approaches, this method aims to ensure that the shocks identified are exogenous, i.e. that they are not influenced by the current state of the economy. The authors use narrative sources (official documents and historical reports) to detect anticipated changes in public spending that are linked to political or military decisions, such as declarations of war or changes in defence budgets. The aim of this method is to capture announcements that influence the expectations of economic agents before the expenditure itself is realised, and which cause the events under consideration in the Granger sense without themselves being caused in the Granger sense (i.e. unidirectionality of predictive capacity). They use an updated version of Ramey's (2011) military news variable.

3.2 Identification of government spending shocks

In order to identify exogenous government spending shocks, the authors use a narrative approach based on defense news³.

These shocks are defined as follows:

$$\text{Shock}_t = \frac{\Delta E_t [G_t^{PDV}]}{\text{GDP}_{t-1}^{\text{nominal}}},$$

²Glass-Steagall Act (1933) securities act, and securities exchange act, for example.

³The major cases of news shocks, which we can see in the graphic, occurred when military spending was announced during the Second World War or the Korean War, but also in the case of announcements of major infrastructure projects, as part of the New Deal, or federal projects in the 1960s. While some significant news shocks have occurred during periods of Slack (the Great Depression and the recession of the 1980s and 2008), these shocks have enabled unused resources to be mobilised. It is important to note, however, that in non-Slack periods, linked to wars, the dynamics illustrated are different, involving the mobilisation of resources for military purposes.

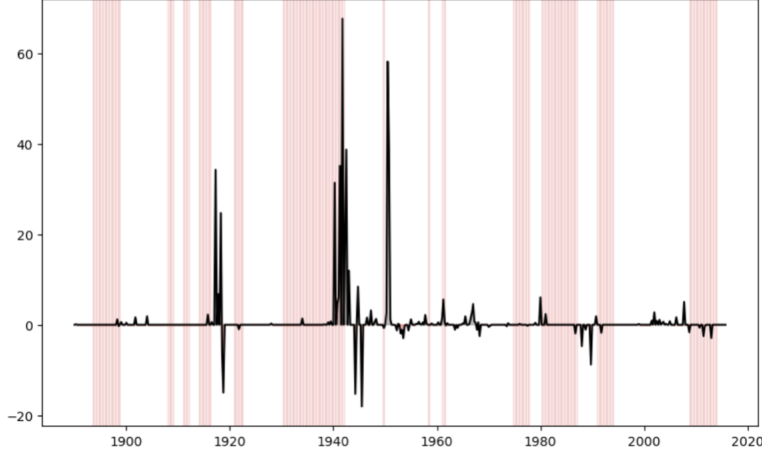


Figure 7: News as percent of GDP and slack periods

where:

- $\Delta E_t [G_t^{PDV}]$ represents the change in the present discounted value of expected future government spending.
- $GDP_{t-1}^{\text{nominal}}$ is the lagged nominal GDP used to normalize the shock.

This ensures that shocks are both exogenous and scaled relative to the size of the economy.

3.3 Local Projection model

The authors employ the **Local Projection method** proposed by Jordà (2005) to estimate impulse responses (IRF). The baseline linear model is specified as:

$$x_{t+h} = \alpha_h + \psi_h(L)y_{t-1} + \beta_h \text{Shock}_t + T_h + \epsilon_{t+h},$$

where:

- x_{t+h} : the variable of interest (e.g., real GDP, government spending) at horizon h .
- y_{t-1} : a vector of control variables, including logs of real per capita GDP, government spending, and defense news.
- $\psi_h(L)$: a lag polynomial of order 2.
- Shock_t : the identified exogenous spending shock.
- T_h : a quartic trend (t^4) to account for long-term trends.
- ϵ_{t+h} : the error term.

The parameter β_h represents the marginal effect of the shock on the variable x_{t+h} , forming the sequence of impulse responses. $\phi(L)$ is the lag polynomial of order 4 (they do not specify whether it has chosen from the minimisation of the an information criterion like AIC or HQ, or from another method e.g. the significance of the lagged last coefficient). For all estimation methods, they use Newey-West corrections to account for serial correlation in residuals, especially for the building of confidence intervals.

3.4 State-dependent model

To capture the dependence of responses on the state of the economy (e.g. periods of slack like recession, vs. of non-slack like expansion), the authors extend the model to allow coefficients to vary by state, for "non-linear" (actually piecewise-linear) LPs:

$$x_{t+h} = I_{t-1} (\alpha_{A,h} + \psi_{A,h}(L)y_{t-1} + \beta_{A,h}\text{Shock}_t) + (1-I_{t-1}) (\alpha_{B,h} + \psi_{B,h}(L)y_{t-1} + \beta_{B,h}\text{Shock}_t) + T_h + \epsilon_{t+h}.$$

With:

- I_{t-1} : a binary indicator for the state of the economy ($I_{t-1} = 1$ if slack, otherwise 0).
- $\alpha_{A,h}, \psi_{A,h}, \beta_{A,h}$: coefficients for periods of slack (A).
- $\alpha_{B,h}, \psi_{B,h}, \beta_{B,h}$: coefficients for non-slack periods (B).

3.5 Dynamic multiplier construction

Instrumental variable (IV) estimation for cumulative multipliers

The cumulative multiplier is derived using the following linear equation for the cumulative response of output (y_{t+j}) over horizon h :

$$\sum_{j=0}^h y_{t+j} = \gamma_h + \phi_h(L)z_{t-1} + m_h \sum_{j=0}^h g_{t+j} + \omega_{t+h},$$

where y_{t+j} represents the GDP at horizon $t+j$, z_{t-1} is a vector of control variables (e.g., lags of output and government spending), g_{t+j} denotes government spending at horizon $t+j$, m_h is the cumulative multiplier at horizon h , $\phi_h(L)$ is a lag polynomial applied to the control variables, and ω_{t+h} is an error term.

To estimate the cumulative multiplier, an Instrumental Variable (IV) technique is used instead of the usual three-step procedure explained *infra*. The sum of government spending shocks, $\sum_{j=0}^h \text{shock}_{t+j}$, is employed as an instrument for the cumulative government spending term, $\sum_{j=0}^h g_{t+j}$, ensuring that the variation in government spending used to identify the multiplier is exogenous. Then, the cumulative multiplier, m_h , can be directly estimated in a single step by regressing the cumulative response of GDP, $\sum_{j=0}^h y_{t+j}$, on the cumulative government spending, $\sum_{j=0}^h g_{t+j}$, using the identified shocks as the instrument.

An equivalent alternative to this efficient and robust one-step IV estimation involves a three-step procedure. First, the response of GDP at each horizon j up to h is estimated, and the cumulative response coefficients are summed to obtain the total GDP effect. Second, the response of government spending at each horizon j up to h is estimated, and the cumulative coefficients are summed to compute the total spending effect. Finally, the multiplier is calculated as the ratio of the cumulative GDP response to the cumulative government spending response. It corresponds to $M_h = \frac{\sum_{i=0}^h \Delta Y_{t+i}}{\sum_{i=0}^h \Delta G_{t+i}}$.

This IV approach has several advantages. First, it allows the multiplier to be estimated directly in a single regression, providing standard errors for the multiplier estimate without requiring separate computations. Second, it ensures that the multiplier is identified using exogenous variation in government spending (the point of an instrument). Finally, by explicitly incorporating the shocks as instruments, it highlights the importance of instrument relevance, which is relevant here with the quest for exogenous shocks.

As they use an instrument, the authors run a weak instrument test (as a weak first-stage would lead to biases). They use the usual 'rule of thumb' value of the F -test for instruments, which is 10. However, this heuristic assumes an absence of serial correlation, which may further bias the test result (Olea & Pflueger, 2013). Thus they also use the corrected F -test and thresholds of the latter.

Finally, for parameters significance and p-values generation, they used heteroscedastic - and autocorrelation - consistent (HAC) standard errors when the instruments are strong, and Anderson and Rubin (1949) ones when they are weak (however, the latter have lower power).

3.6 State indicators and definitions

The state of slack is defined using the unemployment rate, with a threshold of 6.5% as the baseline. Alternative thresholds (e.g. 8%) and a Hodrick-Prescott filtered trend are also tested.

Periods of Zero Lower Bound (ZLB) are identified as times when interest rates are near zero. This is determined relative to deviations from the Taylor Rule:

$$i_t = 1 + 1.5\pi_t + 0.5\text{Output Gap},$$

where π_t is the year-over-year inflation rate.

3.7 Comparison with Blanchard-Perotti's shocks, and combined instruments

As an alternative shock identification to narrative approach, the authors use Blanchard-Perotti (2002) approach, which assumes that government spending is predetermined within the quarter (**short-run restriction**) meaning that it does not react contemporaneously to economic conditions or other variables. This assumption allows government spending shocks to be identified as residuals of the government spending equation in the SVAR (taxes, on the other hand, are allowed to react contemporaneously to economic activity as automatic stabilisers). This method is sensitive to measurement errors in historical data but provides qualitatively similar results, with multipliers below 1 and no significant differences across states. Finally, the authors use the combined instruments (defence news shocks and Blanchard-Perotti's shocks).

3.8 Threshold VAR

The author also build a threshold VAR (TVAR), with as reduced form:

$$Y_t = I_{t-1}\Psi_A(L)Y_{t-1} + (1 - I_{t-1})\Psi_B(L)Y_{t-1} + u_t, \quad (1)$$

with I the dummy variable indicating the state of the economy at the time the shock occurs, and $u_t \sim \mathcal{N}(0, \Omega)$. The covariance matrix Ω is defined as a mixture:

$$\Omega = I_{t-1}\Omega_A + (1 - I_{t-1})\Omega_B,$$

and $\Psi(L)$ represents a polynomial of order 4. y_t and g_t in Y_t are Gordon-Krenn transformations of output and government spending.

They identify both military news shocks and a Blanchard-Perotti shock. A Cholesky decomposition is applied.

As the this method is more "trivial" than non-linear LPs and as the estimates are very similar to those obtained using the Jordà method, we do not reproduced it.

3.9 Other robustness tests

To ensure the validity of the results, the authors conduct an extensive set of robustness checks, about data construction, model specification, and assumptions underlying state definitions and multipliers.

1. **Alternative data interpolation methods for historical data:** Proportional Denton Procedure which is used in the baseline model to interpolate annual data into quarterly series, and linear interpolation applied to the pre-WWII data to smooth out potential volatility induced by the Denton method. While interpolation methods slightly affect the magnitude of the multipliers, they do not alter the main conclusions.
2. **Trends in the model:** To account for potential biases due to omitted long-term trends, a quartic trend (t^4) is included in the baseline model to control for slow-moving changes, such as demographics. Without it, the results remain qualitatively similar but slightly lower multipliers are observed.

3. **Variable transformation through potential GDP scaling:** Following Gordon and Krenn (2010), the authors scale all variables by potential GDP to correct for changes in the level of economic activity:

$$x_t^{\text{scaled}} = \frac{x_t}{\text{Potential GDP}_t}.$$

4. **Alternative state thresholds for slack:** To test the sensitivity of the results to the definition of slack states (6.5% unemployment), alternative thresholds are tested, including 8%, to represent periods of more severe slack, and a time-varying threshold is constructed using the Hodrick-Prescott (HP) filter with a smoothing parameter, capturing deviations from long-term trends in the unemployment rate. Indeed, these thresholds may not correspond to historical definitions of sluggishness, particularly in earlier periods when structural unemployment rates were higher. In all cases, the qualitative findings remain consistent, with no significant differences in multipliers across states. However, differences emerge when the Blanchard-Perotti shock is used with a time-varying threshold.

5. **Smooth transition state definitions:** The authors test a smooth transition function $F(z)$ for defining recessionary periods with a logistic form:

$$F(z_t) = \frac{\exp(-\gamma z_t)}{1 + \exp(-\gamma z_t)},$$

where z_t is a centered 7-quarter moving average of normalized GDP growth and γ is set to 3. This approach allows for a gradual transition between states, rather than discrete thresholds. The results show similar multipliers to those obtained using binary state definitions.

6. **Post-WWII sample restriction and excluding rationing periods:** To assess the impact of the sample period, a restricted sample of 1947-2015 (post-WWII) is analyzed, where the state-dependent multipliers become highly volatile due to weaker instrument relevance, but the main conclusions about the average multiplier remain unchanged. Moreover, periods of severe rationing during WWII (1941Q3 to 1945Q4) are excluded from the analysis, to account for potential distortions in spending behavior caused by supply constraints. The exclusion of these periods slightly increases the overall multipliers but does not alter the state-dependent conclusions.
7. **Role of taxation and deficits:** To analyze the effects of financing, the authors include additional controls for taxes and deficits, with average tax rates (τ_t) calculated as the ratio of federal receipts to nominal GDP:

$$\tau_t = \frac{\text{Federal Receipts}_t}{\text{GDP}_t^{\text{nominal}}}.$$

The deficit is included as a share of GDP. The results show that deficits finance a larger share of government spending during slack states, which theoretically should increase the multiplier. However, the estimated multipliers remain consistent across states.

8. **Zero Lower Bound (ZLB) definition:** They test alternative definitions of ZLB states as their period includes 1932Q2-1951Q1 and 2008Q4-2013Q4, identified by sustained deviations from the Taylor Rule:

$$i_t = 1 + 1.5\pi_t + 0.5\text{Output Gap}.$$

An alternative ZLB definition is based on periods when the 3-month Treasury bill rate is below 50 basis points. The results remain robust, showing no evidence of significantly higher multipliers in ZLB states.

9. **Alternative multiplier calculations:** Finally, authors compare different methods for calculating multipliers. The baseline is integral multiplier:

$$M_h = \frac{\sum_{i=0}^h \Delta Y_{t+i}}{\sum_{i=0}^h \Delta G_{t+i}}.$$

The alternative is the peak response multiplier, i.e. the ratio of the peak output response to the initial spending shock, as used in earlier studies (e.g., Blanchard & Perotti, 2002). They argue that the integral multiplier provides a more policy-relevant measure, as it reflects the cumulative economic effect.

Summary of robustness results

Across all robustness checks, the findings of the paper remain consistent. The government spending multipliers are generally below 1, and there is no significant evidence that multipliers differ across states of the economy or at the zero lower bound.

4 Results and Comments

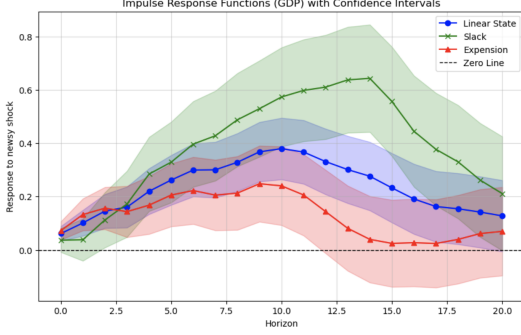


Figure 8: Impulse response for GDP, across states (News shock)

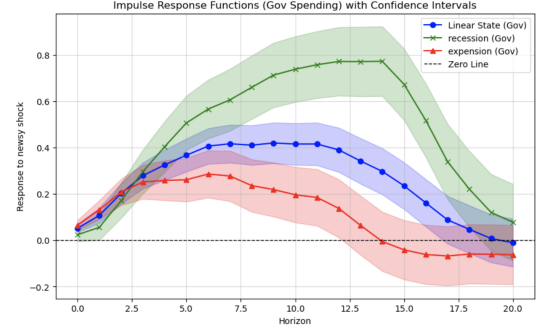


Figure 9: Impulse response for government spending, across states (News shock)

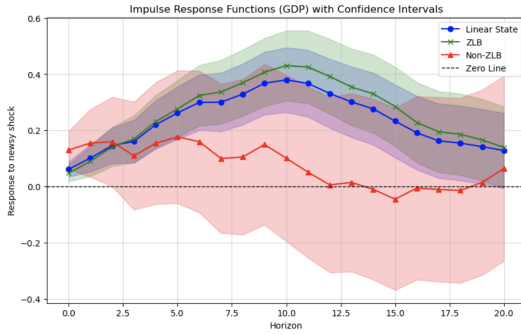


Figure 10: Impulse response of GDP across monetary policy (News shock)

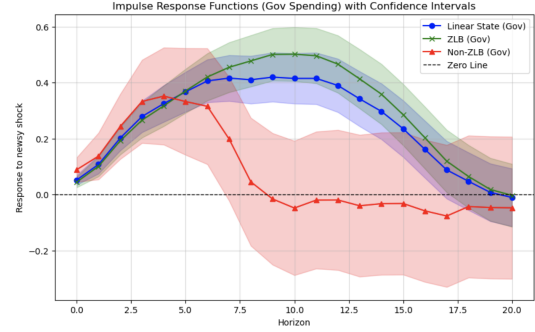


Figure 11: Impulse response for government spending, across monetary policy (News shock)

We present here the results we obtained from our replications.

The impulse response functions seem to indicate that public spending is more effective in slack and monetary constrained environments (Zero lower bound), where fiscal multipliers are amplified. The response is particularly stronger in short and medium horizons (around 10 periods), during recession. During an expansion, the effects are weaker and fade rapidly, potentially reflecting a “crowding-out” effect from public spending to private activity. In the “zero lower bound case”, the response of GDP is stronger and lasts longer. This reflects the amplification of the effect of public spending when there is no compensatory monetary policy to neutralize the shock. In a normal monetary environment, the effects are moderate and fade rapidly. This is consistent with the literature showing that fiscal multipliers are higher in ZLB (Christiano, Eichenbaum, & Rebelo, 2011). We must say that the confidence intervals are relatively high, especially for Figure 10, in the event of a recession. Therefore, these interpretations should be treated with caution.

Although the multipliers generally remain below zero, we note that they are not identical to those in the article, either in terms of multiplier coefficients or confidence intervals and p-values, which are generally higher. The difference between these results and those of the article seems to come from the

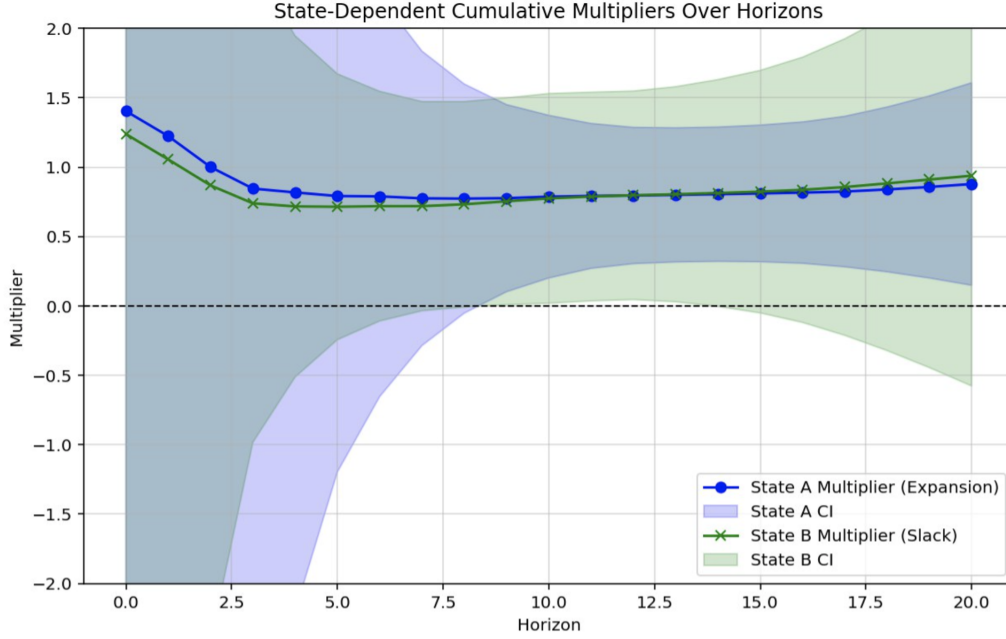


Figure 12: Cumulative spending multipliers across different horizons to a news shock

Table 1: Estimates of Multipliers across states

Shock Type	Horizon	Linear	High unemployment	Low Unemployment	HAC p-value
Military news shock	2-year	0.75 (0.355)	0.75 (0.625)	0.74 (0.312)	HAC = 0.035
	4-year	0.84 (0.435)	0.89 (1.119)	0.82 (0.411)	HAC = 0.053
Blanchard-Perotti shock	2-year	0.50 (0.315)	0.50 (7.335)	0.47 (1.507)	HAC = 0.114
	4-year	0.63 (0.903)	0.64 (7.335)	0.61 (1.507)	HAC = 0.483
Combined	2-year	0.70 (0.237)	0.72 (3.542)	0.68 (0.591)	HAC = 0.003
	4-year	0.81 (0.357)	0.84 (3.542)	0.79 (0.591)	HAC = 0.023

fact that we did not include the Anderson-Rubin correction for weak instruments. This could explain the discrepancy.

4.1 State dependance to fiscal policy, defined by a slack threshold

The most significant shock across the tables is the Combined Shock, due to its statistical robustness (low HAC p values), its consistent multipliers, and its ability to capture both contemporary and anticipatory economic dynamics. Military News Shocks are also interesting but less robust, while Blanchard-Perotti Shocks are clearly the least significant in this context. The multipliers associated with Blanchard-Perotti Shocks are systematically low, around 0.5 to 0.7 at 2 years and 0.6 to 1.0 at 4 years in most configurations. These values are lower than those observed for Military News Shocks and Combined Shocks. Unlike other types of shock, Blanchard-Perotti Shock multipliers show little variation between expansion and recession periods. We also note that multipliers increase slightly with time horizon (from 2 to 4 years), but this progression is modest compared to Military News Shocks and Combined Shocks. The HAC p-values associated with Blanchard-Perotti Shocks are systematically insignificant (≤ 0.1). This means that the differences between low and high unemployment are not statistically robust. These shocks measure immediate variations in public spending or taxes, without taking account of economic agents' expectations.

Comparing tables 1 and 2, it seems that multipliers in these periods remain similar between the two thresholds, with a slight drop in some cases (notably for the Military news shock at 2 years in the table

Table 2: Robustness check: slack threshold 8% unemployment

Shock Type	Horizon	Linear	High unemployment	Low Unemployment	HAC p-value
Military news shock	2-year	0.75 (0.355)	0.74 (0.389)	0.80 (0.431)	HAC = 0.035
	4-year	0.84 (0.435)	0.84 (0.494)	0.82 (0.259)	HAC = 0.053
Blanchard-Perotti shock	2-year	0.50 (0.315)	0.50 (7.335)	0.41 (1.507)	HAC = 0.114
	4-year	0.63 (0.903)	0.64 (7.335)	0.60 (1.507)	HAC = 0.483
Combined	2-year	0.70 (0.237)	0.70 (3.542)	0.63 (0.591)	HAC = 0.003
	4-year	0.81 (0.357)	0.81 (3.542)	0.77 (0.591)	HAC = 0.023

Table 3: Moving Average of Output Growth Weighting Function

Shock Type	Horizon	Low unemployment	Linear	High Unemployment	HAC p-value
Military news shock	2-year	0.75 (0.355)	0.76 (0.796)	0.79 (1.588)	HAC = 0.035
	4-year	0.84 (0.435)	0.86 (0.591)	0.88 (1.078)	HAC = 0.053
Blanchard-Perotti shock	2-year	0.50 (0.315)	0.51 (7.335)	0.41 (1.507)	HAC = 0.114
	4-year	0.63 (0.903)	0.64 (7.335)	0.55 (1.507)	HAC = 0.483
Combined	2-year	0.70 (0.237)	0.64 (3.542)	0.54 (0.591)	HAC = 0.003
	4-year	0.81 (0.357)	0.77 (3.542)	0.69 (0.591)	HAC = 0.023

with an 8% threshold, just as in the original article) for the slack state. This relative stability could reflect the fact that multipliers in slack periods are fairly robust to an increase in the unemployment threshold. By increasing the unemployment threshold from 6% to 8%, fewer spells are classified as “High Unemployment” spells. This may lead to the inclusion of periods with weaker or more variable multiplier effects. Indeed, slack periods defined by an 8% threshold often include contexts where the economy is already severely disrupted or in severe recession. In such cases, factors such as declining confidence among economic agents⁴ or additional market frictions⁵ (such as financial constraints or low labor supply elasticity) can reduce the effectiveness of public spending, even in periods of high unemployment. Economic agents may also anticipate that increased public spending will be followed by a future increase in taxes to finance deficits, which reduces their propensity to consume or invest today (Romer & Romer, 2010)

⁴In periods of very high unemployment (above 8%), the economy is often in deep recession or suffers severe shocks, which can lead to a significant drop in the confidence of economic agents (households and businesses). This can translate, for example, into a drop in the propensity to consume (Eggertsson & Krugman, 2012). Businesses, meanwhile, are likely to cut back on investment in the face of a gloomy economic outlook, even if demand is temporarily boosted by public spending (Hall, 2009).

⁵When unemployment is high, frictions such as skill mismatch or worker discouragement can limit the labor supply response to stimulus measures (Michaillat, 2014). Long-term workers, faced with prolonged unemployment, may exit the labor market or lose their skills, making the labor supply response more rigid (Krueger and Mueller, 2012).

Table 4: Robustness check: period 1947-2015

Shock Type	Horizon	Low unemployment	Linear	High Unemployment	HAC p-value
Military news shock	2-year	0.63 (1.068)	0.92 (12.338)	0.72 (3.785)	HAC = 0.556
	4-year	0.57 (3.512)	0.92 (11.031)	0.70 (4.239)	HAC = 0.872
Blanchard-Perotti shock	2-year	0.36 (1.012)	0.37 (29.433)	0.33 (8.828)	HAC = 0.725
	4-year	0.39 (5.164)	0.38 (29.433)	0.36 (8.828)	HAC = 0.940
Combined	2-year	0.61 (0.932)	0.66 (20.806)	0.62 (6.248)	HAC = 0.513
	4-year	0.56 (3.152)	0.59 (20.806)	0.57 (6.248)	HAC = 0.860

Table 5: Estimates of Multipliers excluding WWII

Shock Type	Horizon	Low unemployment	Linear	High Unemployment	HAC p-value
Military news shock	2-year	1.33 (4.828)	1.37 (5.867)	2.06 (14.884)	HAC = 0.783
	4-year	1.69 (6.993)	1.73 (12.292)	2.46 (22.795)	HAC = 0.809
Blanchard-Perotti shock	2-year	0.69 (1.110)	0.68 (6.378)	0.66 (4.423)	HAC = 0.536
	4-year	1.04 (3.064)	1.03 (6.378)	1.02 (4.423)	HAC = 0.735
Combined	2-year	1.17 (2.611)	1.17 (8.721)	1.17 (6.089)	HAC = 0.655
	4-year	1.54 (4.309)	1.54 (8.721)	1.56 (6.089)	HAC = 0.721

For the table using the moving average to define slack periods, we note that, as in the article, the results are relatively counter-intuitive. Multipliers are higher in periods of low unemployment than in periods of high unemployment, for all types of shocks and horizons. The p-values show significant results for Combined Shocks, but not for Blanchard-Perrotti shocks, indicating that multipliers in periods of low and high unemployment do not differ robustly. This seems to reflect their contemporary, rather than anticipatory, nature.

For non-World War II estimates, we note the particularly high confidence interval, making it impossible to interpret.

4.1.1 A small note on weak instruments

The author, running F -tests and corrected-for-auto-correlation F -tests for weak instruments, find that defense news shocks are weak instruments. We do not reproduce those tests but it must be taken into consideration to assess the validity of point estimates, as weak instruments lead to bias.

4.2 Interaction between monetary policy and fiscal policy

Tables 6 and 7 confirm that multipliers are significantly higher in ZLB, particularly for Military News Shocks and Combined Shocks. These results underline the importance of monetary constraint in the effectiveness of fiscal policies, with effects amplified when monetary policy is constrained. Blanchard-Perotti shocks, on the other hand, remain weak and insensitive to these variations, confirming their contemporary and less dynamic nature. Multipliers are systematically higher in the ZLB than in a normal monetary environment. The p-values show marginal significance for these results ($p < 0.05$). For the Blanchard-Perotti shocks, the results remain insignificant ($p \geq 0.1$). instruments.

In ZLB, nominal interest rates cannot be cut any further to compensate for the increase in demand induced by public spending. This amplifies fiscal multipliers, as monetary policy does not intervene to neutralize the fiscal impact. In Table 7, where the ZLB is strictly defined ($T\text{-Bill} \leq 0.5\%$), this effect is even more pronounced (1.07 at 4 years). This is in line with the findings of Christiano, Eichenbaum, & Rebelo (2011). The low multipliers and low sensitivity to the ZLB reflect the direct nature of Blanchard-Perotti shocks. These shocks do not trigger amplifying mechanisms linked to agents' expectations, which limits their impact in different monetary contexts. The lack of variation between the ZLB and the normal scenario is consistent with the literature showing that the direct effects of public spending are not strongly dependent on the monetary framework.

Once again, the most significant shocks remain the combined. In ZLB, the combined multipliers increase significantly, exceeding 1.0 at 4 years in Table 7. This shows that the interaction between fiscal and monetary policy is crucial: when monetary policy is constrained (ZLB), public spending becomes a more powerful tool for stimulating the economy.

News shocks are particularly weak as instruments, especially for the post-WWII period. Thus, we should be really careful when considering the obtained point estimates.

Table 6: Estimates of Multipliers accross monetary policy

Shock Type	Horizon	Normal case	Zero lower bound	linear	HAC p-value
Military news shock	2-year	0.75 (0.355)	0.62 (1.821)	0.71 (0.719)	HAC = 0.035
	4-year	0.84 (0.435)	0.77 (6.008)	0.82 (2.296)	HAC = 0.053
Blanchard-Perotti shock	2-year	0.50 (0.315)	0.52 (7.335)	0.50 (1.507)	HAC = 0.114
	4-year	0.63 (0.903)	0.63 (7.335)	0.63 (1.507)	HAC = 0.483
Combined	2-year	0.70 (0.237)	0.77 (3.542)	0.72 (0.591)	HAC = 0.003
	4-year	0.81 (0.357)	0.87 (3.542)	0.83 (0.591)	HAC = 0.023

Table 7: Defining ZLB as T-Bill Rate ≤ 0.5

Shock Type	Horizon	Normal case	Zero lower bound	linear	HAC p-value
Military news shock	2-year	0.75 (0.355)	0.87 (2.791)	0.76 (0.944)	HAC = 0.035
	4-year	0.84 (0.435)	1.07 (4.913)	0.87 (1.549)	HAC = 0.053
Blanchard-Perotti shock	2-year	0.50 (0.315)	0.58 (7.335)	0.51 (1.507)	HAC = 0.114
	4-year	0.63 (0.903)	0.76 (7.335)	0.66 (1.507)	HAC = 0.483
Combined	2-year	0.70 (0.237)	0.85 (3.542)	0.72 (0.591)	HAC = 0.003
	4-year	0.81 (0.357)	1.01 (3.542)	0.84 (0.591)	HAC = 0.023

Table 8: MP multiplier excluding WWII

Shock Type	Horizon	Normal case	Zero lower bound	linear	HAC p-value
Military news shock	2-year	1.33 (4.828)	0.69 (1.679)	1.07 (1.917)	HAC = 0.783
	4-year	1.69 (6.993)	0.89 (4.960)	1.24 (4.647)	HAC = 0.809
Blanchard-Perotti shock	2-year	0.69 (1.110)	0.71 (6.378)	0.83 (4.423)	HAC = 0.536
	4-year	1.04 (3.064)	1.12 (6.378)	1.34 (4.423)	HAC = 0.735
Combined	2-year	1.17 (2.611)	0.96 (8.721)	1.13 (6.089)	HAC = 0.655
	4-year	1.54 (4.309)	1.15 (8.721)	1.38 (6.089)	HAC = 0.721

5 Discussion

5.1 External validity and generalisability

As the paper focuses on US historical data, the results may not be generalisable to other economies with different institutional structures, fiscal capacities and monetary policy frameworks. For example, emerging economies or economies subject to significant fiscal constraints might exhibit higher multipliers, particularly in periods of slowdown or crisis; European countries subject to strict fiscal rules, such as those bound by the Stability and Growth Pact, might experience different dynamics due to the limited flexibility of their fiscal policy; or economies with fixed exchange rates or dollarised systems, with their limited monetary policy responses and where fiscal policy is the main stabilisation tool.

5.2 Multiplier definition

In addition, they consider that the relevant multipliers to consider are those cumulative over 2 and 4 years, whereas short-term peak multipliers may also be relevant for public decision-makers in the event of a crisis, as studied in particular by Blanchard-Perotti.

5.3 Linear-piecewise construction of state-dependence: choice of thresholds, and definition and dynamics of states

While the general specification for LPs is said to be non-linear (NL-LP), in reality it is piecewise linear through the indicator function based on the transition variable, in the manner of a Threshold VAR (TVAR). This has several consequences for the specification:

1. ZLB states are defined on the basis of Treasury bill rates and deviations from the Taylor rule, but the choice of 50 basis points or specific historical periods may omit nuances in monetary policy regimes (for example, periods of quasi-ZLB with forward guidance).
2. Assuming that state transitions (e.g. from an eased to a non-eased state) follow average historical patterns, with linearity within each state, is also problematic. However, during extreme periods (e.g. the Great Depression or the Great Recession), these transitions can deviate significantly from historical norms, so that multipliers can vary non-linearly according to the intensity of slack or the depth of a recession. Recession and the COVID-19 pandemic may exhibit higher multipliers given the size of the trough and prolonged ZLB conditions. The author used a smooth transition specification (with a logistic transition function) and find no significant differences in results. However, we can still imagine that fiscal responses to war may differ systematically from peacetime interventions, which connects with the issue of the shock definition.
3. The analysis separates periods of slack and ZLB states, but does not take account of their interaction, even though they often coincide, making it difficult to distinguish the distinct effects of each state. The joint effects of a high degree of leeway and accommodative monetary policy could differ from their individual impacts.
4. Finally, the fact that multipliers do not vary from one state to another may also be partly due to averaging effects: the heterogeneity of shocks (e.g. defence spending in wartime versus fiscal responses to recessions in peacetime) may dilute the state-dependent dynamics.

We could model state transition probabilities as time-varying or non-linear, with a regime-switching model (e.g. Markov-switching regime) based on a Kalman filter and a smoother. The advantage of this method is that it does not rely on arbitrary criteria (e.g. NBER Business Cycle Dating Committee) or on ‘natural’ trend hypotheses (natural output or unemployment) to identify regime changes: a filter updates the probability of state at time t conditional on the information at time t , and the smoother then updates the probability of state in a backward manner conditional on all the information obtained in $t+1$ a posteriori (Hamilton algorithm). We could also introduce interaction terms to jointly analyse spells and ZLB states (for example, a combined dummy variable for high unemployment and ZLB spells).

5.4 Identifying structural shocks and focusing on defence spending

The use of the narrative approach to identify structural shocks exogenous to the policy makers' response function to the economic situation can be questioned here. Firstly, the strategy of taking news shocks to military spending (linked to going to war) as exogenous tax expenditure naturally leads to testing only the impact of public spending mainly associated with military spending, which is particularly targeted, sectoral and non-fungible, and may therefore have very different transmission mechanisms to other spending shocks such as public works programmes or social transfers, which are not taken into account even though they represent major components of public spending, particularly during slack periods (e.g. New Deal programmes during the recession). g. New Deal programmes during the Great Depression). In fact, defence shocks can produce dynamics that are not typical of fiscal policy, including at the aggregate level through possible passthroughs.

Although the argument in defence of news shocks about defence spending is that geopolitical events are indeed much more exogenous than countercyclical policy reactions and other automatic stabilisers, including in terms of Granger causality, the fact remains that other narrative shocks could have been used to identify a slightly more representative impact than military spending from broader categories of public spending. For example, the method used by Romer and Romer (2010), which reconstructs a series of shocks from presidential speeches and congressional reports in order to identify tax policies implemented for reasons that are orthogonal to a pure reaction to the economic situation and its prospects.

The second limitation concerns the purpose of the news shocks themselves. Narrative series are generally considered as time series of structural shocks and are directly integrated into models such as LP, SVAR, ADRL... But these series are not, strictly speaking, the shocks we are interested in. They contain a lot of noise (measurement error), and are also subject to the effects of anticipation or other contemporary fiscal policies that could blur the identification of the impacts of public spending. They are therefore variables that are plausibly correlated with a particular structural shock and uncorrelated with others. These narrative shocks should therefore be treated as external instruments, which is the case in the article. As a consequence, they rule weak instrument tests. News shocks are particularly weak as instruments, especially for the post-WWII period.

Finally, the validity of defence shocks as exogenous instruments is not explicitly tested in the paper. They may not be orthogonal to economic conditions, particularly in periods of uncertainty: wars and geopolitical crises may coincide with wider economic disturbances affecting public spending decisions beyond defence.

5.5 Local projections method

Although very flexible, LPs have known limitations. At longer horizons, IRFs estimated by LPs tend to become erratic due to noise and the limited number of data points (they are less efficient than SVARs), and the use of Newey-West standard errors may not fully correct for long-horizon autocorrelation, leading to even less reliable confidence intervals. The results may also lack the robustness of methods with dynamic identification restrictions such as SVARs. We could compare the results of local projections with those obtained from SVAR or Bayesian VARs, and we could use the block bootstrap method to calculate confidence intervals for long horizons.

5.6 Temporal inconsistencies in data construction

The construction of key variables, such as real GDP, government procurement and unemployment rates, combines multiple data sources over time, as post-WWII data is based on modern statistical methodologies, while pre-WWII data is based on reconstructed historical series, leading to potential inconsistencies in definitions and measurements. For example, the concept of GDP was not formally defined until the 1940s, and earlier series may have a different composition (for example, by excluding certain services or government activities). In addition, the article does not take full account of the measurement errors inherent in historical data, particularly for earlier periods (e.g. measurement of the unemployment rate). Furthermore, the Denton proportional interpolation method used for data prior to the Second World War introduces potential biases, as this method assumes that quarterly trends are smooth. Finally, data relating to public spending prior to the Second World War has been reconstructed and may therefore only be partial, particularly with regard to non-defence spending.

Finally, even in the case where the measurement error is ‘random’ (white noise, with no systematic bias), it will still lead to an attenuation bias, i.e. causing the result to tend towards 0, due to a non-informative variance.

5.7 Exclusion of impacts unrelated to GDP

The focus is naturally on the reaction of GDP to public spending, which is the definition of the multiplier. However, public spending can also affect other macroeconomic variables of interest (employment and unemployment rates, inflation and price dynamics (particularly in the ZLB, where monetary policy is limited), private sector investment and consumption (potential crowding-out or crowding-in effects) and distributional outcomes such as inequality or sectoral differences. There are many channels through which fiscal policy can have an impact, and these are not taken into account solely through the prism of the multiplier - all the more so if we take the economic slack of employment as a state transition variable (spending multipliers could manifest themselves mainly through employment gains during periods of slack, even if the response of GDP is attenuated, or fiscal policy could stabilise prices during periods of deflation at the ZLB).

5.8 No discussion of transmission channels

Finally, the article is purely empirical and a-theoretical, and in fact makes no mention of the mechanisms behind the multipliers observed (crowding-out effects? Lack of responsiveness on the part of the private sector? Negative anticipation on the part of households? Sectoral allocation of spending, in this case in the military sphere). These analyses will also have consequences for the design of optimal public policies.

6 References

Anderson, T. W., & Rubin, H. (1949). Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations.” *Annals of Mathematical Statistics*. 20:46–63.

Auerbach, A. J., & Gorodnichenko, Y. (2012). Measuring the Output Responses to Fiscal Policy. *American Economic Journal: Economic Policy*, 4(2), 1–27. <https://doi.org/10.1257/pol.4.2.1>

Auerbach, A. J., & Gorodnichenko, Y. (2013). Fiscal Multipliers in Recession and Expansion. In *Fiscal Policy after the Financial Crisis* (NBER Chapters). National Bureau of Economic Research. <https://doi.org/10.3386/w17447>

Barro, R. J. (1981). Output effects of government purchases. *Journal of Political Economy*.

Blanchard, O., & Perotti, R. (2002). An Empirical Characterization of the Dynamic Effects of Changes in Government Spending and Taxes on Output. *The Quarterly Journal of Economics*, 117(4), 1329–1368. <https://doi.org/10.1162/003355302320935043>

Christiano, L. J., Eichenbaum, M., & Rebelo, S. (2011). When Is the Government Spending Multiplier Large? *Journal of Political Economy*, 119(1), 78–121. <https://doi.org/10.1086/659312>

Crafts, N., & Mills, T. C. (2013). Fiscal Policy in a Depressed Economy: Was There a ‘Free Lunch’ in 1930s’ Britain? *The Economic History Review*, 66(4), 1120–1143. <https://doi.org/10.1111/j.1468-0289.2012.00672.x>

Eggertsson, G. B. (2011). What Fiscal Policy is Effective at Zero Interest Rates? *NBER Macroeconomics Annual 2010*, 25, 59–112. <https://doi.org/10.1086/652407>

Eggertsson, G. B., & Krugman, P. (2012). Debt, deleveraging, and the liquidity trap: A Fisher-Minsky-Koo approach. *The Quarterly Journal of Economics*.

Gordon, R. J., & Krenn, R. (2010). The End of the Great Depression 1939–41: Policy Contributions and Fiscal Multipliers. *NBER Working Paper Series*, No. 16380. <https://doi.org/10.3386/w16380>

Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.

Krueger, A. B., & Mueller, A. (2012). The lot of the unemployed: A time use perspective. *Journal of the European Economic Association*

Jordà, Ò. (2005). Estimation and Inference of Impulse Responses by Local Projections. *American Economic Review*, 95(1), 161–182. <https://doi.org/10.1257/0002828053828518>

Michaillat, P. (2014). A Theory of Countercyclical Government Multiplier. *American Economic Journal: Macroeconomics*.

Olea, J. L. M., & Pflueger, C. (2013). A Robust Test for Weak Instruments. *Journal of Business & Economic Statistics*, 31(4), 358–369. <https://doi.org/10.1080/07350015.2013.771027>

Ramey, V. A. (2011). Identifying Government Spending Shocks: It’s All in the Timing. *Quarterly Journal of Economics*, 126 (1): 51–102. <https://doi.org/10.1093/qje/qjq008>

Ramey, V. A., & Shapiro, M. D. (1998). Costly Capital Reallocation and the Effects of Government Spending. *Carnegie-Rochester Conference Series on Public Policy*, 48, 145–194. [https://doi.org/10.1016/S0167-2231\(98\)00020-7](https://doi.org/10.1016/S0167-2231(98)00020-7)

Ramey, V. A., & Zubairy, S. (2017). Government Spending Multipliers in Good Times and in Bad: Evidence from US Historical Data. *Journal of Political Economy*, 125(2), 850–901. <https://doi.org/10.1086/691094>

Romer, C. D., & Romer, D. H. (1983). The Role of Fiscal Policy in Stabilization Policy. *NBER Working Paper Series*, No. 0845. <https://doi.org/10.3386/w0845>

Romer, C. D., & Romer, D. H. (2010). The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks. *American Economic Review*, 100(3), 763–801. <https://doi.org/10.1257/aer.100.3.763>

Samuelson, P. A. (1939). Interaction Between the Multiplier Analysis and the Principle of Acceleration. *The Review of Economics and Statistics*, 21(2), 75–78. <https://doi.org/10.2307/1927758>

Tinbergen, J. (1939). *Statistical Testing of Business Cycle Theories*. League of Nations.

Fazzari, S. M., Morley, J., & Panovska, I. (2014b). State-dependent effects of fiscal policy. *Studies In Nonlinear Dynamics And Econometrics*, 19(3), 285–315. <https://doi.org/10.1515/snde-2014-0022>